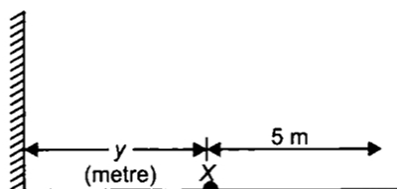


LIGHT -01

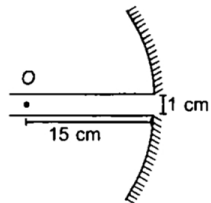
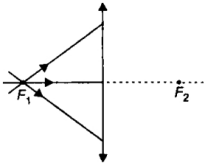
Class 10 - Science

1. If parallel beams, non-parallel to principal axis fall on the convex lens, they converge at a point: [1]
 - a) away from principal axis
 - b) called focus on the axis
 - c) on principal axis
 - d) centre of curvature
2. A man is standing at a position X in front of a plane mirror, a distance of y metre from the mirror as shown in figure. When the man moves 5 m away from the mirror, the new distance between the man and his image becomes 20 metre. What is the value of y? [1]



- a) 5 m
 - b) 40 m
 - c) 10 m
 - d) 20 m
3. A ray passing through the focus and falling on a convex lens will: [1]
 - a) retrace its path
 - b) will emerge parallel to the principal axis
 - c) will emerge through a focus on another side
 - d) will emerge perpendicular to the principal axis
4. Magnification produced by a rear view mirror fitted in vehicles [1]
 - a) is equal to one
 - b) can be more than or less than one depending upon the position of the object in front of it
 - c) is less than one
 - d) is more than one
5. A ray of light passes from glass into air. The angle of refraction will be: [1]
 - a) greater than the angle of incidence
 - b) equal to the angle of incidence
 - c) smaller than the angle of incidence
 - d) 45 degrees
6. A student has to do the experiment on finding the focal length of a given convex lens by using a distant object. She can do her experiment if she is also made available with [1]
 - a) a lamp and a screen
 - b) a scale and a screen
 - c) a lamp and a scale
 - d) None of these
7. A piece of red cloth when suitably illuminated may look black, but a piece of black cloth will never appear red. This phenomenon occurs because: [1]

- 2 / 10

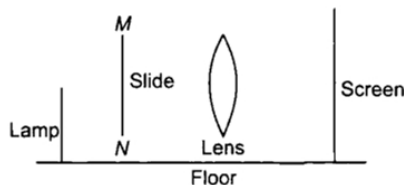
- c) A concave lens of focal length 50 cm d) A concave lens of focal length 5 cm
18. How many images are seen when a ray of light gets reflected by two plane mirrors which are set up at an unknown angle between them that the ray undergoes a deviation of 200° ? [1]
- a) 4 b) 5
c) 3 d) 2
19. The image formed by a concave mirror is real, inverted, and of the same size as the object. The position of the object must then be: [1]
- a) beyond the centre of curvature b) at the centre of curvature
c) at the focus d) between the centre of curvature and focus
20. A mirror is broken into two parts and these parts are separated by a distance of 1 cm as shown in figure. The focal length of the mirror is 10 cm. The height of the images (in cm with respect to their optic axis), formed by the two parts of mirror when the object is midway between the two principal axes are [1]
- 
- a) 2,2 b) 0.5,1
c) 1,2 d) 1,1
21. In the given diagram, three rays of light pass through focal point F_1 to the converging lens, placed along vertical line. Which of the following is correct regarding this? [1]
- 
- a) The speed of light rays remains same while entering or leaving the converging lens. b) The speed of light rays decreases when they enter the converging lens.
c) The speed of light rays increases when they enter the converging lens. d) The speed of light rays decreases when they leave the converging lens.
22. An object is 100 mm in front of a concave mirror which produces an upright image (erect image). The radius of curvature of the mirror is: [1]
- a) less than 100 mm b) more than 200 mm
c) between 100 mm and 200 mm d) exactly 200 mm
23. Your school laboratory has one large window. To find the focal length of a concave mirror using one of the walls as the screen, the experiment may be performed: [1]
- a) on the wall adjacent to the window b) on the same walls as the window
c) only on the table as per the laboratory arrangement d) near the wall opposite to the window
24. The focal length of a spherical mirror of radius of curvature 30cm is: [1]

- a) 30cm
b) 15cm
c) 20cm
d) 10cm

25. An achromatic lens has a focal length of 10 cm. At what distance should the mirror from the lens be settling so that it forms an image at 20 cm from the lens and also calculate the amplification produced by the achromatic lens? [1]

- a) $u = 1.25$ cm and $m = 12.05$ cm
b) $u = -0.02$ cm and $m = 20$ cm
c) $u = 20$ cm and $m = 10$ cm
d) $u = -1.05$ cm and $m = 19.05$ cm

26. In a slide projector, light from a lamp is incident on to a slide MN and a lens is used to form an image of the slide on a screen as shown in figure. What kind of image is formed on the screen? [1]



- a) Real, magnified, inverted
b) Virtual, diminished, erect
c) Virtual, magnified, erect
d) Real, diminished, inverted

27. The focal length of a concave mirror whose radius of curvature is 32cm: [1]

- a) None of these
b) 32 cm
c) 12 cm
d) 16 cm

28. A student obtains a blurred image of an object on a screen by using a concave mirror. In order to obtain a sharp image on the screen, he will have to shift the mirror. [1]

- a) towards the screen
b) to a position very far away from the screen
c) depending upon the position of the object
d) away from the screen

29. While positioning the pins, it is arranged such that [1]

- a) the foots are in perpendicular line
b) the heads are in the straight line
c) the mid-points are in the straight line
d) the foots are in straight line

30. The real image formed by a concave mirror is smaller than the object if the object is: [1]

- a) at a distance equal to radius of curvature
b) at a distance greater than radius of curvature
c) at a distance equal to focal length
d) between centre of curvature and focus

31. The refractive indices of four substances P, Q, R, and S are 1.50, 1.36, 1.77, and 1.31 respectively. The speed of light is the maximum in the substance: [1]

- a) R
b) S
c) P
d) Q

32. The image of a distant object is obtained on a screen by using a concave mirror. The focal length of the mirror can be determined by measuring the distance between [1]

- a) the mirror and the screen
b) the object and the screen
c) the object and the mirror
d) None of these

33. An object is placed in front of a convex mirror at infinity. According to the New Cartesian Sign Convention, the [1]

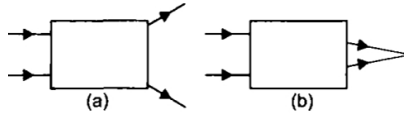
sign of the focal length and the sign of the image distance in this case are respectively:

- a) +, -
b) +, +
c) -, -
d) -, +

34. When an object is placed at a distance of 15 cm from a concave mirror, its image is formed at 10 cm in front of the mirror. The focal length of the mirror: [1]

- a) 10 cm
b) 4 cm
c) 8 cm
d) 6 cm

35. In the following figures, optical instruments are placed inside boxes (a) and (b). Which of the following options is correct for these? [1]



- a) In (b), the incident rays are converged after refraction, so lens is concave.
b) In (a), the incident rays are diverged after coming out of the box, so mirror is convex.
c) In (a), the incident rays are diverged after refractions, so lens is concave.
d) In (b), the incident rays are converged after coming out of the box, so mirror is convex.

36. The nature of the image formed by the concave mirror when the object is placed between C and F is: [1]

- a) Virtual and erect
b) Real and Virtual
c) Real and Inverted
d) Real and erect

37. The refractive indices of four materials A, B, C, and D are 1.33, 1.43, 1.71, and 1.52 respectively. When the light rays pass from air into these materials, they refract the maximum in: [1]

- a) material B
b) material A
c) material D
d) material C

38. The refractive index (n) of a colourless, volatile, non-polar solvent is 1.24 to the refractive index of a solid ice cube. The unmitigated refractive index of the ice cube is 1.31. Enumerate the unmitigated refractive index of the non-polar solvent. [1]

- a) 1.6522
b) 1.6244
c) 1.3241
d) 1.2401

39. If the angle of incidence is increased for a pair of air - glass interface, then the angle of refraction will [1]

- a) increase
b) remains the same
c) decrease
d) first increases and then decreases

40. Parallel rays from a distant tree incident on a convex lens from an image of the screen. The diagram correctly showing the image of the tree on the screen is: [1]

- a)
b) None of these



41. A man stands 10 m in front of a large plane mirror. How far must he walk before he is 5 m away from his image? [1]

a) 8.5 m b) 8.0 m
c) 7.0 m d) 7.5 m

42. An object is kept at a distance more than twice the focal length (F) from a concave mirror. The image will be formed at a distance: [1]

a) Between F and $2F$ b) Equal to F
c) More than $2F$ d) Less than F

43. The refractive indexes of four substances P, Q, R, and S are 1.77, 1.50, 2.42, and 1.31 respectively. When light travelling in air is incident on these substances at equal angles, the angle of refraction will be the maximum in: [1]

a) substance S b) substance P
c) substance Q d) substance R

44. A ray of light travelling in water falls at right angles to the boundary of a parallel-sided glass block, the ray of light: [1]

a) is refracted away from the normal b) is reflected along the same path
c) does not get refracted d) is refracted towards the normal

45. On which of the following, the efficient working of optical fibre works upon? [1]

a) Scattering of light b) Reflection of light
c) Refraction of light d) Total internal reflection

46. Refraction cannot cause bending as light moves from one surface to another if the incident and refraction angles i and r are related as: [1]

a) $i = r = 90^\circ$ b) $i = 0^\circ = r = 90^\circ$
c) $i \neq r = 0^\circ$ d) $i = r = 0^\circ$

47. An achromatic glass used for end-vision on a car with a radius of 4.00 m. If a car is situated at 6.00 m from this achromatic glass, find the position of the image. [1]

- a) 4 m
b) 3 m
c) 5 m
d) 2 m

48. A student wants to obtain an erect image of an object using a concave mirror of 10 cm focal length. What will be the distance of the object from mirror? **[1]**
a) Less than 10 cm
b) 10 cm
c) more than 20 cm
d) between 10 cm and 20 cm

49. Light travelling from a denser medium to a rarer medium along a normal to the boundary: **[1]**
a) goes along the bounding
b) is refracted away from the normal
c) is refracted towards the normal
d) is not refracted

50. A ray of light is incident on a plane mirror making an angle of 90 degree with the mirror surface. The angle of reflection for this ray of light will be: **[1]**
a) 60 degree
b) 90 degree
c) 45 degree
d) 0 degree

51. To determine the focal length of a concave mirror, a student focuses a distant object using the concave mirror. The best object can be: **[1]**
a) a distant tree
b) All of these
c) classroom window
d) sun

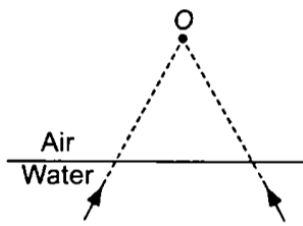
52. As the incident angle is increased for a given pair of the medium, the refraction angle will **[1]**
a) decrease
b) remain the same
c) zero
d) increase

53. A spherical mirror and a thin spherical lens have each a focal length of -15 cm. The mirror and the lens are likely to be **[1]**
a) The mirror is convex, but the lens is concave.
b) Both convex
c) Both concave
d) The mirror is concave and the lens is convex.

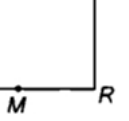
54. In a convex spherical mirror, reflection of light takes place at: **[1]**
a) a bulging-out surface
b) a bent-in surface
c) an uneven surface
d) a flat surface

55. The image of an object formed by a plane mirror is: **[1]**
a) virtual
b) upside-down
c) real
d) diminished

56. The image formed by a concave mirror is observed to be virtual, erect and larger than the object. Where should be the position of the object? **[1]**
a) Between the principal focus and the centre of curvature
b) Beyond the centre of curvature

- c) At the centre of curvature
d) Between the pole of the mirror and its principal focus.
57. The refractive index of water with respect to air is $\frac{4}{3}$. The refractive index of air with respect to water will be: [1]
a) 0.50
b) 0.75
c) 0.25
d) 1.75
58. In torch lights and head lights of vehicles, the bulb is placed [1]
a) between the pole and the focus of the reflector.
b) very near to the focus of the reflector.
c) at the centre of curvature of the reflector.
d) between the focus and centre of curvature of the reflector.
59. The image of an extended object placed in front of a concave mirror is formed at a distance of 40 cm from the object. If the image is 3 times bigger than the object, the magnitude of focal length of the mirror is [1]
a) 5 cm
b) 10 cm
c) 20 cm
d) 15 cm
60. **Statement 1:** Even when one half of a convex lens is covered with a black paper, the lens will produce a complete image. [1]
Statement 2: Intensity of image is reduced when we cover the half of a convex lens with a black paper.
a) Statement 1 is true but statement 2 is false.
b) Both statements 1 and 2 are false.
c) Both statements 1 and 2 are true but statement 2 is not the correct explanation of statement 1.
d) Both statements 1 and 2 are true and statement 2 is the correct explanation of statement 1.
61. A converging set of rays, travelling from water (with refractive index $\frac{4}{3}$ to air) is incident on a plane interface, the rays would have converged to a point 0,40 cm above the interface. However, due to refraction the rays will bend. At what distance above the interface will the rays actually converge? [1]
- 
- a) 30 cm
b) 53.3 cm
c) 25 cm
d) 10 cm
62. The laws of reflection hold true for: [1]
a) convex mirrors only
b) concave mirrors only
c) all reflecting surfaces
d) plane mirrors only
63. If two lenses of power P_1 and P_2 are put in contact, what will be the net power? [1]
a) $P_1 \times P_2$
b) $P_1 - P_2$
c) $\frac{P_1}{P_2}$
d) $P_1 + P_2$

64. PQRS shows the walls of room each measuring 8 m, such that wall PQ has a full length plane mirror. A camera is placed at point M, the mid point of wall SR. In order to take clear image of the point R, the camera should be focused at. [1]



a) More than 16 metres b) 16 m
 c) 8 m d) 4 m

65. The speed of light, in a given medium, is $\frac{2}{3}$ rd of its speed in a vacuum. The absolute refractive index of the medium is equal to: [1]

a) $\frac{3}{2}$ b) $\frac{4}{9}$
 c) $\frac{2}{3}$ d) $\frac{9}{4}$

66. Rays from Sun converge at a point 15 cm in front of a concave mirror. Where should an object be placed so that size of its image is equal to the size of the object? [1]

a) 30 cm in front of the mirror b) between 15 cm and 30 cm in front of the mirror
 c) more than 30 cm in front of the mirror d) 15 cm in front of the mirror

67. Which of the following phenomena is not the result of total internal reflection? [1]

a) Looming b) Sparkles of the diamond
 c) Mirage d) Twinkling of stars

68. Where should an object be placed in front of a convex lens to get a real image of the size of the object? [1]

a) At 2F b) At Infinity
 c) At focus d) Between the optical centre and the focus

69. The laws of reflection hold good for [1]

a) convex mirror only b) concave mirror only
 c) plane mirror only d) all mirrors irrespective of their shape

70. If R is the radius of curvature of a spherical mirror and f is its focal length then: [1]

a) $R = f$ b) $R = \frac{f}{2}$
 c) $R = 3f$ d) $R = 2f$

71. When a ray of light from rarer to denser it will: [1]

a) Return into the same medium b) Bend away from the normal
 c) Pass straight d) Bends towards the normal

72. When the two opposite surfaces of the slab are not parallel, the emergent angle e and incident angle are related as [1]

a) $e < i$ b) $e > i$

c) does not depend on the refractive index of the medium

d) depends on the refractive index of the medium

73. The image formed by a concave mirror is virtual, erect, and magnitude. The position of object is: [1]

a) between pole and focus

b) between focus and centre of curvature

c) at focus

d) at pole

74. A lens of focal length ' f ' is cut into two equal parts without affecting its curvature. The two pieces will have equal focal length of : [1]

a) $\frac{f}{2}$

b) f

c) $\frac{f}{3}$

d) $2f$

75. The formula which gives the relationship between the object distance, image distance and focal length of the spherical mirror is: [1]

a) $\frac{1}{v} = \frac{1}{u} + \frac{1}{f}$

b) $\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$

c) $\frac{1}{v} + \frac{1}{u} = \frac{1}{f}$

d) $\frac{1}{v} = \frac{1}{u} + \frac{1}{f}$